

National Health and Nutrition Examination Survey

1999-2023 Data Documentation, Codebook, and Frequencies

Dietary Supplement Database - Product Information (DSPI)

Data File: DSPI.xpt

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Component Description

The NHANES Dietary Supplement Database (NHANES-DSD) contains detailed information on the dietary supplements (DS) and non-prescription antacids containing calcium and/or magnesium (antacids) reported by survey participants since NHANES 1999. The NHANES-DSD release consists of three datasets containing information on the following: Dietary Supplement Product Information (DSPI), Dietary Supplement Ingredient Information (DSII), and Dietary Supplement Blend Information (DSBI).

Dietary supplement information from the in-house NCHS Product Label Database (PLD) is publicly released in three files that make up the NHANES-DSD. These files incorporate all products that have been reported by respondents since 1999 from the PLD. With subsequent releases, new products reported will be appended to the NHANES-DSD files.

The in-house PLD database is maintained by NCHS nutritionists. NCHS attempts to obtain a product label for all dietary supplements or antacids reported by participants from sources such as the manufacturer or retailer, the Internet, company catalogs, and the Physician's Desk Reference (PDR). Label information has also been obtained from the [Dietary Supplement Label Database \(DSLDB\)](#), which is a joint project of the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) and National Library of Medicine (NLM). The DSLDB contains the full label contents from a sample of dietary supplement products marketed in the U.S. Selected label information is then entered into the PLD including, but not limited to: supplement name; manufacturer and/or distributor; serving size; form of serving size; and ingredients and amounts. The ingredient information entered into the database is taken directly from the supplement facts box on the label or carton. The PLD is used to assist with data editing and then publicly released in the data files (DSPI, DSII, and DSBI).

In addition to entering labels into the PLD database, NCHS created generic and default dietary supplements and antacids, which are also maintained in the database. Generics were created in the database when we had information about a reported supplement, such as the strength of all ingredients, but no brand name. These were generally single ingredient supplements, which included a strength (e.g., vitamin C 500 mg) or multiple vitamins and/or mineral supplements made by a private label manufacturer that was known to us and for which we had a label with identical ingredients and strengths (e.g., brand X all-purpose multivitamin was reported, and we had a label for brand Y, made by the same manufacturer). When all ingredient strengths were unknown, a default supplement was created in the database. Defaults were created for single ingredient and multiple ingredient supplements based on our own data of most frequently reported supplements of that type, based on the survey cycle in which the data was collected. Created default products and the actual products or strengths that were assigned to these defaults are listed in the documentation for the files associated with participants' use of dietary supplements, located on the [NHANES website](#) under the dietary data links.

In 2019, the application used to manage and access NHANES dietary supplement product database was updated to adapt to the .NET environment. This resulted in the new structure of dietary supplement identifiers. Variables DSDPID, DSDIID, DSDIBID were added to indicate the updated supplement ID, ingredient ID, and blend component ID, respectively. The variables

DSDSUPID, DSDINGID, and DSDBCID now indicate the old versions for supplement ID, ingredient ID, and blend component ID, respectively. In addition, six new variables were included to provide information on supplement reformulations: DSDPRDT, DSDPREID, DSDORGID, DSDSGPF, DSDSEQF, DSDLINRF to indicate the product type, previous product ID, original product ID, sequential group formulation, sequential formulation, and linear formulation, respectively.

Data Processing and Editing

The in-house PLD was used for processing and editing of the dietary supplement data since 1999.

Variables pulled directly from the in-house PLD

DSDSUPP: Name of Supplement

This is the name from the front of the product label. The brand name is generally entered first (i.e., Nature's Way) and then the actual product name (i.e., lutein). Information such as the strength (i.e., 60 mg extract) of the product and other qualifiers that help distinguish a product from a similar product (i.e., mega, super, high potency, time release, chewable, extract) are also listed if they are on the front of the product label. Words like "dietary supplement" and "health claims" are not entered as part of the name.

DSDPID: Supplement ID Number

New Supplement ID is a unique number assigned to each product entered in the inhouse PLD.

DSDPRDT: Product Type

If the supplement was entered by NCHS as a regular product, it is equal to 1. If the supplement was created by NCHS as a generic product, it is equal to 2. Otherwise, if the supplement was created by NCHS as a default product, it is equal to 3. Previous Variable DSDGENRC was replaced by code "2" in DSDPRDT.

DSDSUPID: Supplement ID Number-Old version

Old Supplement ID is a 10-digit identifier assigned to each product entered into the in-house PLD. All identifiable products have a supplement ID beginning with the number "1". The next 3 digits (positions 2-4) are "888" or "881" if the supplement was created by NCHS as a generic or default product; otherwise the digits in positions 2-4 are coded "000" or "001". The next 4 digits (positions 5-8) are sequentially assigned by the system for each product. The last 2 digits (positions 9-10) indicate formulations of the same supplement: the first formulation entered into the database = 00, the first reformulation = 01, the next = 02, etc. Note that these are reformulations of the same product. Different versions of products (e.g., liquid vs. tablet, with iron vs. without iron, regular vs. high potency) are considered as different products thus have different 4-digit numbers in positions 5 to 8. When a product name was entered as "no product information available," "refused," or "don't know," its ID number starts with a string of 6s.

DSDSRCE: Supplement Information Source

The source of each product label is recorded into the PLD. Generic and default products do not have a source code.

DSDTYPE: Supplement Type

The type of supplements is recorded into the database. The supplement types are: infant/pediatric, prenatal, mature, or standard . Products are coded as infant/pediatric when the product name states "infant, children, child, or kid/s" or has an indication in the title, label, or the form (e.g., animal shapes) of the supplement that it is intended for children. If this is not the case, but the suggested dose or directions indicate dosage for children only, then the code is infant/pediatric, but if dosages for adults are also included, then the product is coded as standard. Products are coded as "Prenatal" when the product name states prenatal or a derivative of this name or has an indication in the title or label that it is intended for pregnant women. If this is not the case, but the suggested dosage or directions indicate dosage for pregnant women only, the product is coded as a prenatal, but if dosages for non-pregnant adults are also included, the product is coded as standard. Products are coded as "mature"

when the product name or label includes words such as "mature, senior, geriatric, post-menopausal, or silver" or indicates with other words that it is intended for individuals 50 years and over. All other products are coded as "standard".

DSDSERVQ: Serving Size Quantity

This is the "serving size quantity," which is recorded from the product label's supplement facts panel for which the nutrient amounts are based on.

DSDSERVU: Serving Size Unit

This is the "serving size unit," which is recorded from the product label's supplement facts panel.

DSDPREID: Previous product ID

This represents the parent product from which this current product was derived.

DSDORGID: Original product ID

This represents the original or root product from which this formulation or product was derived.

DSDSGPF: Sequential group formulation

This represents the sequential order of creation from the original product within the group of identical product types.

DSDSEQF: Sequential formulation

This represents the sequential order of creation from the original product.

DSDLINRF: Linear formulation

This represents the direct line of descendancy order of creation from the original product. For example- original product being level 1, a product derived from original level 1 product being level 2 and a product derived from the level 2 product being level 3, and so on.

DSDIID: Ingredient ID

This is the new unique ingredient ID created by the PLD for each ingredient recorded from the product label's supplement facts panel.

DSDINGID: Ingredient ID-Old version

This is the old unique ingredient ID created by the PLD for each ingredient recorded from the product label's supplement facts panel.

DSDINGR: Ingredient Name

Ingredient names are recorded from the product label's supplement facts panel.

DSDOPER: Ingredient Operator

This is a symbol =, <, or > that comes from the product label's supplement facts panel.

DSDQTY: Ingredient Quantity

Ingredient quantity is recorded for each ingredient listed from the product label's supplement facts panel. Some nutrient amounts are for a nutrient compound (generally a foreign-made product or an antacid) and these must be converted to an elemental nutrient amount (See Appendix 2 for recommended conversions).

DSDUNIT: Ingredient Unit

Ingredient unit is recorded for each ingredient listed from the product label's supplement facts panel.

NHANES dietary supplement files have vitamin A and E expressed in different units based on what was reported on the product label's supplement facts panel. Vitamin E may be reported in International Units (IU) or micrograms (µg). Vitamin A may be reported as International Units

(IU) or as micrograms (μg) of retinol activity equivalents (RAE). Users should be aware of changes in dietary supplement labeling and are advised to use appropriate unit conversions, [Unit Conversions](#).

DSDCAT: Ingredient Category

There are ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

DSDBCNAM: Blend Component Name

These are the ingredient names found within a blend. Blends in products will not give the actual breakdown of ingredient quantities in the blend. The ingredients will usually just be listed, and most of the time a total blend amount is given. The blend ingredients are released in file DSBI.

DSDBID: Blend Component ID

These are new unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCLID: Blend Component ID-Old version

These are old unique ingredient ID numbers for blend ingredients created by the PLD.

DSDBCCAT: Blend Component Category

There are blend ingredient categories: Vitamin, Mineral, Botanical, Others, Amino Acid. These are assigned by NCHS staff. (Please see Appendix 1: Rules for Classifying Ingredients.)

Variables created for the NHANES-DSD public release files**DSDBLFLG: Blend Flag**

This indicator variable denotes whether an ingredient is a blend or not a blend. If the ingredient is a blend, then blend ingredients are contained in file DSBI.

DSDCNTV: Number of Vitamin(s) in Supplement

This variable indicates the number of vitamins in the supplement, including in blends. All ingredients categorized as "vitamins" in variables DSDCAT and DSDBCCAT are added up to get the number of vitamins in the product. If a product has the same vitamin listed as an ingredient as well as a blend, this vitamin was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTM: Number of Mineral(s) in Supplement

This variable indicates the number of minerals in the supplement, including in blends. All ingredients categorized as "minerals" in variables DSDCAT and DSDBCCAT are added up to get the number of minerals in the product. If a product has the same mineral listed as an ingredient as well as a blend, this mineral was only counted once. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTA: Number of Amino Acid(s) in Supplement

This variable indicates the number of amino acids in the supplement, including in blends. All ingredients categorized as "amino acids" in variables DSDCAT and DSDBCCAT are added up to get the number of amino acids in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTB: Number of Botanical(s) in Supplement

This variable indicates the number of botanicals in the supplement, including in blends. All ingredients categorized as "botanicals" in variables DSDCAT and DSDBCCAT are added up to get the number of botanicals in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

DSDCNTO: Number of Other Ingredients in Supplement

This variable indicates the number of other ingredients in the supplement, including in blends. All ingredients categorized as "other" in variables DSDCAT and DSDBCCAT are added up to get the number of other ingredients in the product. Ingredients that are categorized as a blend (DSDBLFLG=1) in file DSII are not included in this count, but the ingredients within the blend are counted (blend ingredients in file DSBI).

The NCHS-DSD file may be linked to both the Household dietary supplement data use files and the 24-hour dietary recall interview dietary supplement files by the supplement ID number (DSDPID).

NCHS-DSD contains information on all DS and antacids reported from 1999-2020. New products will be appended as they are reported in future data releases. There are three files. The "Supplement Information" file (DSPI) and the "Ingredient Information" file (DSII) can be linked by supplement ID number (DSDPID), which is a unique product identifier. The "Ingredient Information" file (DSII) and the "Supplement Blend" file (DSBI) can then be linked by the ingredient ID number (DSDIID).

Layout of DSQ_DSD**(DSPI): Supplement Product Information**

Variable Name	Label
DSDPID	Supplement ID number
DSDPRDT	Product Type
DSDSUPP	Supplement Name
DSDSRCE	Supplement Information Source
DSDTYPE	Supplement Type
DSDSERVQ	Serving Size Quantity
DSDSERVU	Serving Size Unit
DSDPREID	Previous product ID
DSDORGID	Original product ID
DSDSGPF	Sequential group formulation
DSDSEQF	Sequential formulation
DSDLINRF	Linear formulation
DSDCNTV	Count of Vitamins in the Supplement
DSDCNTM	Count of Minerals in the Supplement
DSDCNTA	Count of Amino Acids in the Supplement
DSDCNTB	Count of Botanicals in the Supplement
DSDCNTO	Count of Other Ingredients in the Supplement
DSDSUPID	Supplement ID number-Old version

(DSII): Supplement Ingredient Information

Variable Name	Label
DSDPID	Supplement ID Number
DSDSUPP	Supplement Name
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDOPER	Ingredient Operator (<, =, >)
DSDQTY	Ingredient Quantity
DSDUNIT	Ingredient Unit
DSDCAT	Ingredient Category
DSDLFLG	Blend Flag
DSDINGID	Ingredient ID Number- Old version

(DSBI): Supplement Blend Information

Variable Name	Label
DSDIID	Ingredient ID Number
DSDINGR	Ingredient Name
DSDDBID	Blend Component ID Number
DSDBCNAM	Blend Component Name
DSDBCCAT	Blend Component Category
DSDBCID	Blend Component ID Number-Old version

Analytic Notes**Source of Product Information**

The best source of product information is the label itself, but when this cannot be obtained, other sources are used. Information from other sources may not always be an accurate

reflection of what is actually on the supplement label. This is true for the supplement's apparent name as well as for the ingredients. The apparent name on the container is most important, since interviewers see the supplement container and record the name as it appears to them. Differences from what appears on the label are particularly noted for information from the Internet (name and ingredients), the PDR (name), and supplement carton (name). In addition, supplement companies may change the appearance of a label and thus the apparent name without changing the content or may change content with minimal change to the label, or may change both. NCHS attempts to obtain updated labels as they come onto the market, but cannot guarantee complete success. The source of the supplement information is included in the data release.

Using Self-Reported Data

NHANES data are self-reported and recorded by interviewers, and thus may contain inconsistencies or errors. Some inconsistencies have been edited; however, users may notice situations that still need editing. Users are advised to assess the data and edit it as deemed appropriate for the analyses being undertaken.

Codebook and Frequencies

DSDPID - SUPPLEMENT ID NUMBER

Variable Name: DSDPID

SAS Label: SUPPLEMENT ID NUMBER

English Text: Supplement ID number

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28553	Range of Values	20235	20235	
.	Missing	0	20235	

DSDPRDT - PRODUCT TYPE

Variable Name: DSDPRDT
SAS Label: PRODUCT TYPE
English Text: Product type

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Regular	17885	17885	
2	Generic	1513	19398	
3	Default	837	20235	
.	Missing	0	20235	

DSDSUPP - SUPPLEMENT NAME

Variable Name: DSDSUPP
SAS Label: SUPPLEMENT NAME
English Text: Supplement Name

Code or Value	Value Description	Count	Cumulative	Skip to Item
SUPPLEMENT NAME	Value was recorded	20235	20235	
< blank >	Missing	0	20235	

DSDSRCE - SUPPLEMENT INFORMATION SOURCE

Variable Name: DSDSRCE

SAS Label: SUPPLEMENT INFORMATION SOURCE

English Text: Supplement Information Source

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Directly from manufacturer	23	23	
2	Directly from distributor	7	30	
4	Inferred from supplement name	1	31	
5	Physician's Desk Reference (PDR)	31	62	
7	Product Catalog	149	211	
8	Internet Listing	4812	5023	
9	Supplement label or carton	11750	16773	
10	Supplement from same manufacturer	143	16916	
11	Dietary Supplement Label Database	1013	17929	
.	Missing	2306	20235	

DSDTYPE - SUPPLEMENT TYPE

Variable Name: DSDTYPE

SAS Label: SUPPLEMENT TYPE

English Text: Supplement Type (prenatal, infant\pediatric, standard, or mature)

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Infant/pediatric	1172	1172	
2	Prenatal	407	1579	
3	Mature	509	2088	
4	Standard	18147	20235	
.	Missing	0	20235	

DSDSERVQ - SERVING SIZE QUANTITY

Variable Name: DSDSERVQ
SAS Label: SERVING SIZE QUANTITY
English Text: Serving size quantity

Code or Value	Value Description	Count	Cumulative	Skip to Item
0.02 to 60	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSERVU - SERVING SIZE UNIT

Variable Name:	DSDSERVU
SAS Label:	SERVING SIZE UNIT
English Text:	Serving size unit

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Caplet	1033	1033	
2	Capsule	4995	6028	
3	Dropper	59	6087	
4	Drop	132	6219	
5	Fluid Ounce	99	6318	
6	Gel Cap	25	6343	
9	Lozenge	49	6392	
10	Milliliter	314	6706	
12	Package/Package	412	7118	
13	Pill	51	7169	
14	Tablespoon/Powder	106	7275	
16	Softgel	2889	10164	
17	Tablespoon/Liquid	203	10367	
18	Tablet	7096	17463	
19	Teaspoon/Liquid	267	17730	
20	Wafer	47	17777	
21	Ounce/Powder	4	17781	
22	Spray/Squirt	23	17804	
24	Scoop/Powder	410	18214	
25	Cup/Powder	8	18222	
27	Chew	1135	19357	
29	Vegicap	477	19834	
30	Can/Liquid	2	19836	
31	Capful	39	19875	
32	Gumball	3	19878	
33	Gram/Powder	94	19972	
34	Teaspoon/Powder	225	20197	
37	Cup/Liquid	1	20198	
38	Gram/Liquid	1	20199	
39	Drop/Lozenge	9	20208	
42	Strip	2	20210	
43	Bottle/Liquid	12	20222	
44	Pieces	6	20228	
99	Unknown	7	20235	
.	Missing	0	20235	

DSDPREID - PREVIOUS PRODUCT ID

Variable Name: DSDPREID
SAS Label: PREVIOUS PRODUCT ID
English Text: Previous product ID

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28550	Range of Values	5842	5842	
.	Missing	14393	20235	

DSDORGID - ORIGINAL PRODUCT ID

Variable Name: DSDORGID
SAS Label: ORIGINAL PRODUCT ID
English Text: Original product ID

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 28550	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSGPF - SEQUENTIAL GROUP FORMULATION

Variable Name: DSDSGPF

SAS Label: SEQUENTIAL GROUP FORMULATION

English Text: Sequential group formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 19	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSEQF - SEQUENTIAL FORMULATION

Variable Name: DSDSEQF
SAS Label: SEQUENTIAL FORMULATION
English Text: Sequential formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 29	Range of Values	20235	20235	
.	Missing	0	20235	

DSDLINRF - LINEAR FORMULATION

Variable Name: DSDLINRF

SAS Label: LINEAR FORMULATION

English Text: Linear formulation

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 12	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTV - COUNT OF VITAMINS IN THE SUPPLEMENT

Variable Name: DSDCNTV

SAS Label: COUNT OF VITAMINS IN THE SUPPLEMENT

English Text: Count of vitamins in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 18	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTM - COUNT OF MINERALS IN THE SUPPLEMENT

Variable Name: DSDCNTM

SAS Label: COUNT OF MINERALS IN THE SUPPLEMENT

English Text: Count of minerals in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 72	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTA - COUNT OF AMINO ACIDS IN THE SUPPLEMENT

Variable Name: DSDCNTA

SAS Label: COUNT OF AMINO ACIDS IN THE SUPPLEMENT

English Text: Count of amino acids in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 24	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTB - COUNT OF BOTANICALS IN THE SUPPLEMENT

Variable Name: DSDCNTB

SAS Label: COUNT OF BOTANICALS IN THE SUPPLEMENT

English Text: Count of botanicals in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 125	Range of Values	20235	20235	
.	Missing	0	20235	

DSDCNTO - COUNT OF OTHER INGREDIENTS IN SUPPLEMENT

Variable Name: DSDCNTO

SAS Label: COUNT OF OTHER INGREDIENTS IN SUPPLEMENT

English Text: Count of other ingredients in the supplement, including those in blends

Code or Value	Value Description	Count	Cumulative	Skip to Item
0 to 53	Range of Values	20235	20235	
.	Missing	0	20235	

DSDSUPID - SUPPLEMENT ID NUMBER - OLD VERSION**Variable Name:** DSDSUPID**SAS Label:** SUPPLEMENT ID NUMBER - OLD VERSION**English Text:** Supplement ID Number - old version

Code or Value	Value Description	Count	Cumulative	Skip to Item
SUPPLEMENT ID NUMBER - OLD VERSION	Value was recorded	14629	14629	
< blank >	Missing	5606	20235	

Appendix 1: Rules for Classifying Ingredients

VITAMINS

An ingredient is classified as a vitamin if it is:

- A single vitamin listed by its name (e.g., vitamin A)
- A standard chemical form of the vitamin (retinol, retinal, retinoic acid) in synthetic or natural form

A vitamin will be classified as "other" when it exists as:

- A precursor or provitamin to the active form of the vitamin (e.g., 7-dehydrocholesterol, a precursor to Vitamin D)
- A complex, since the ingredient content is unclear (e.g., B-complex)

The following appear in supplements as a source of vitamins and are therefore classified as a vitamin:

- Vitamin A: palmitate, vitamin A acetate, vitamin A palmitate
- Vitamin B-1/Thiamin: thiamin monophosphate or TMP, thiamin mononitrate, thiamin hydrochloride
- Vitamin B-2/Riboflavin: riboflavin mononitrate, riboflavin-5-phosphate sodium
- Vitamin B-3/Niacin
- Vitamin B-5/Pantothenic Acid: pantothenate, calcium pantothenate
- Vitamin B-6: pyridoxine hydrochloride, vitamin B6 hydrochloride
- Vitamin B-12/Cobalamin: cyanocobalamin
- Vitamin C/Ascorbic Acid: ascorbyl palmitate, sodium ascorbate
- Vitamin D/Calciferol: cholecalciferol, ergocalciferol, calcitriol
- Vitamin E/Tocopherol: d-alpha tocopheryl acid succinate, dl-alpha tocopheryl acetate, d-alpha tocopheryl acetate, d-alpha tocopherol, d-alpha tocopheryl, tocopherols, mixed tocopherols, vitamin E acetate, tocotrienol
- Vitamin K/Menadione: phytonadione
- Biotin: Choline, choline bitartrate
- Folic Acid/Folate

MINERALS

An ingredient is classified as a mineral if it is a macro or micromineral (trace element):

- in its elemental form (e.g., iron)
- is the source of the mineral in a supplement (e.g., ferrous gluconate, potassium iodide, nickel chloride)

An ingredient containing a mineral is classified as "other" when it is:

- an enzyme (e.g., boron protease)
- a complex, since the ingredient content is unclear (e.g., Trace Mineral Complex) used as an electrolyte (chloride, potassium, sodium)

The following are classified as minerals:

- Arsenic
- Copper
- Phosphorus

- Barium
- Fluoride
- Selenium
- Boron
- Iodine
- Silicon
- Bromine
- Iron
- Strontium
- Cadmium
- Magnesium
- Sulfur
- Calcium
- Manganese
- Tin
- Chromium
- Molybdenum
- Vanadium
- Cobalt
- Nickel
- Zinc

BOTANICALS

An ingredient is classified as a botanical if it is:

- part of a plant, tree, shrub, herb, etc.

Botanicals may include the following words:

- Extract, Powder
- Leaf, Root, Flower, Stem, Peel, Fruit
- Component of a botanical that specifically mentions it is from the plant (e.g., soy isoflavones, citrus bioflavonoids)

An ingredient containing a botanical is classified as "other" if it is:

- listed only as an unspecified blend
- a chemical structure derived originally from a botanical (e.g., bromelain, the enzyme found in pineapple; Alliin, a phytochemical in garlic; apple cider vinegar)

AMINO ACIDS

An ingredient is classified as an amino acid if it is an essential or nonessential amino acid. It can exist in:

- its free form (e.g., lysine, glutamine)
- its post-translational form with one or two added groups (e.g., Cystine, Hydroxylysine, Hydroxyproline, Dimethylglycine, and 3-methylhistidine)
- one of its isomeric forms (e.g., L-tyrosine)
- the source of an amino acid in a supplement (e.g., L-lysine monohydrochloride, glutamic acid hydrochloride)

An amino acid would be classified as "other" if it is:

- in its post-translational form with three or more added groups (Trimethylglycine, Tetramethylglycine, etc.)
- an alpha-keto acid (an amino acid with its amino group, NH₃, replaced by a keto group) (e.g., -ketoglutarate)
- a residue of an amino acid (-carboxyglutamic acid also known as GLA)
- as a complex of amino acids (e.g., natural amino acid complex), since the ingredient content is unclear

The following are classified as amino acids:

- Alanine
- Glycine
- Proline
- Arginine
- Histidine
- Serine
- Asparagine
- Isoleucine
- Taurine
- Aspartic
- Acid
- Leucine
- Threonine
- Cysteine
- Lysine
- Tryptophan
- Glutamic
- Acid
- Methionine
- Tyrosine
- Glutamine
- Phenylalanine
- Valine

OTHER

The following are examples of ingredients that would be classified as "other":

- an electrolyte (e.g., chloride, potassium, sodium)
- a hormone (e.g., DHEA, cholesterol), unless if it is the active form of a vitamin (calcitriol)
- an enzyme (e.g., cellulase, glucoamylase)
- Complexes and blends (unless all components are of the same type ex. amino acid blend)
- Bioflavonoids and Isoflavones (if not associated with a plant, in which case it would be classified as a Botanical)
- Vinegars
- Phytochemicals (e.g., lutein, allin)
- Vitamin precursors (e.g., some carotenoids)

Appendix 2: Conversion Factors for Supplement Nutrient Units to Food Units and for Nutrient Compounds to Elemental Nutrients

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
Vitamin A Conversion Factors		
ALPHA CAROTENE	587	1 IU alpha carotene = 7.2 mcg vitamin A
ALPHA CAROTENE	587	1 RAE = 24 mcg alpha carotene
BETA CAROTENE	392	1 IU beta carotene = 0.6 mcg vitamin A
BETA CAROTENE	392	1667 IU beta carotene = 1 mg beta carotene
BETA CAROTENE	392	1 RAE = 12 mcg beta carotene
VITAMIN A*	360	1 IU = 0.3 mcg vitamin A
VITAMIN A*	360	1 RAE = 1 mcg vitamin A
CRYPTOXANTHIN	614	1 RAE = 24 mcg cryptoxanthin
Vitamin D Conversion Factor		
VITAMIN D†	364	40 IU vitamin D = 1 mcg
Vitamin E Conversion Factor		
VITAMIN E‡	365	1 IU = 0.67 mg vitamin E
Calcium Conversion Factors		
CALCIUM CARBONATE	543	40% elemental calcium
CALCIUM L-THREONATE	4003	12.9 % elemental calcium
CALCIUM PANTOTHENATE	395	91.6% pantothenate
Iron Conversion Factors		
FERROUS FUMARATE	782	32.9% elemental iron
IRON FERROCHEL	13380	27.65% elemental iron
Glucosamine Conversion Factors		
GLUCOSAMINE HYDROCHLORIDE	410	83.0% glucosamine
GLUCOSAMINE SULFATE	143	65% glucosamine
GLUCOSAMINE SULFATE .2 KCL	850	29.6% glucosamine
D-GLUCOSAMINE SULFATE.2 NACL	1017	31.3% glucosamine
Magnesium Conversion Factors		
MAGNESIUM ASPARTATE	13643	8.42% elemental magnesium
MAGNESIUM CARBONATE	556	28.9% elemental magnesium
MAGNESIUM HYDROXIDE	544	41.7% elemental magnesium
MAGNESIUM PHOSPHATE TRIBASIC	616	27.7% elemental magnesium
MAGNESIUM TRISILICATE	2054	18.3 % elemental magnesium
Vitamin B-6 Conversion Factor		
PYRIDOXINE HYDROCHLORIDE	470	82% vitamin B-6
Other		
ALUMINUM HYDROXIDE GEL	14221	34.59% elemental aluminum

INGREDIENT	INGREDIENT_ID	CONVERSION FACTOR
CHROMIUM NICOTINATE	14270	12.43% elemental chromium
CHROMIUM PICOLINATE	487	12.4% elemental chromium
CHOLINE BITARATE	82	41% choline
CHOLINE CITRATE	2248	41% choline
CREATINE MONOHYDRATE	480	88% creatine
CYSTEINE HCL	776	76.9% cysteine
DOCUSATE SODIUM	109	5.1% sodium
GLUTAMIC ACID HYDROCHLORIDE	653	80.1% glutamic acid
L-CYSTEINE HCL	488	69.0% cysteine
L-GLUTAMIC ACID HCL	611	80.1% glutamic acid
L-LYSINE HCL	743	80.03% lysine
LYSINE HYDROCHLORIDE	2088	80.03% lysine
POTASSIUM CHLORIDE	287	52.5% elemental potassium
POTASSIUM PHOSPHATE	575	28.7% elemental potassium
POTASSIUM PHOSPHATE MONOBASIC	615	28.7% elemental potassium
THIAMIN MONONITRATE	468	92% thiamin
ZINC PICOLINATE	2629	21.1% elemental zinc
Basic Unit Conversion		
1 gm = 1000 mg		
1 mg = 1000 mcg		

* Conversion factor used for Vitamin A is Retinol, most common form

† Conversion factor for Calciferol

‡ Conversion factor for Alpha Tocopherol, most common form